

MODULE : TARGET POPULATION, SAMPLING TECHNIQUES, AND DATA COLLECTION METHODS

INSTRUCTIONAL HOURS: 4

1 MODULE OVERVIEW

In this module, you will explore sampling strategies, data collection methods, and their appropriateness for different types of research in data science.

2 LEARNING OUTCOMES

By the end of this module, you will be able to:



1. Explain the concepts of population and sampling in research
2. Select appropriate sampling techniques and tools for effective data collection.
3. Design a sampling strategy tailored to a specific research question.
4. Apply ethical considerations and inclusivity in participant selection and data collection procedures.

3 LEARNING ACTIVITIES/ TASK LIST

1. View the curated lecture video
2. Study assigned notes and Quiz
3. Design a mini-project plan
4. Complete participation a real-world case study

4 INSTRUCTIONAL MATERIALS & LEARNING ACTIVITIES

LEARNING ACTIVITY 1: READING NOTES AND ASSIGNMENTS

Read the following notes and answer the quiz that follow.

1.UNDERSTANDING TARGET POPULATION IN RESEARCH

The **target population** in research refers to the entire set of individuals, data points, or events that are of interest to a study. In the realm of data science, this might involve a wide range of subjects such as users of a digital platform, patients in a hospital, consumers of a specific product, or entries in a dataset related to public health trends or financial transactions. The goal is to identify a group that is relevant to the research questions and objectives being pursued.

To better understand how to work with populations, it is necessary to become familiar with several related terms:

- **Population:** This includes all elements; people, transactions, records, events; that share one or more characteristics which the researcher intends to study. It is the full group about which conclusions are to be drawn.
- **Accessible Population:** This is the portion of the target population that the researcher can realistically reach or contact. Limitations such as geography, budget, and access restrictions often necessitate narrowing the population scope.
- **Sampling Frame:** This refers to the actual list, database, or mechanism used to identify and draw the sample. An accurate and updated sampling frame is essential to ensure the representativeness of the selected sample.
- **Sample:** A subset of the population selected for analysis.
- **Sampling Unit:** The individual element (e.g., person, household, data point) selected for study.
- **Bias:** Systematic deviation of sample results from true population values.

- **Precision:** The degree to which sample estimates reflect the true population value.

Example: Suppose a data scientist is investigating fraudulent activities in online banking. The **target population** could be all bank transactions conducted in a specific year. However, the **accessible population** might only include those transactions that are available through a bank's internal monitoring system. The **sampling frame** would be the list of transactions flagged by the bank's fraud detection software.

PRINCIPLES OF SAMPLING

Sampling refers to the process of selecting a manageable subset from a larger population, with the intent of making inferences about the whole. In data science, this practice is vital due to the massive volumes of data often encountered, which make full population analysis impractical.

Importance of Sampling

Sampling allows researchers to conduct studies more efficiently, using fewer resources while still obtaining meaningful insights. Moreover, well-designed samples enable statistical analyses that can be generalized to the broader population with known levels of confidence and error.

Characteristics of a High-Quality Sample

A good sample should meet the following criteria:

- **Representativeness:** The sample should accurately reflect the characteristics and diversity of the overall population.
- **Unbiased Selection:** Every unit should have an equal or known chance of being included, to avoid systematic distortion.
- **Sufficient Size:** The number of units sampled should be adequate to support the desired statistical analysis and ensure robust conclusions.

Failure to meet these conditions can result in **sampling bias**, leading to skewed results that do not truly represent the target population.

SAMPLING TECHNIQUES

Sampling methods can broadly be divided into **probability** and **non-probability** sampling. The choice between them depends on the research purpose, desired level of statistical precision, and available resources.

A. Probability Sampling Methods

In probability sampling, each member of the population has a known, non-zero chance of being selected. These methods support statistical generalization to the population.

1. Simple Random Sampling:

Every member of the population has an equal chance of being chosen. This method is straightforward and requires a complete and accessible sampling frame. For instance, selecting 100 customer records at random from a database of 10,000.

2. Stratified Sampling:

The population is divided into mutually exclusive groups (strata) based on shared characteristics like age, income, or geographic location. A random sample is drawn from each stratum, ensuring all segments are proportionally represented. This approach is useful for improving precision and ensuring key subgroups are not overlooked.

3. Systematic Sampling:

A starting point is chosen at random, and every k-th element is selected thereafter. This technique is efficient when dealing with a list or ordered dataset, such as selecting every 10th log entry in a system database.

Formula: $k = \text{Population Size (N)} / \text{Sample Size (n)}$

4. Cluster Sampling:

The population is divided into clusters (often based on geography or institutions), and entire clusters are randomly selected. This method is

beneficial when the population is widely dispersed, as it reduces data collection costs.

5. **Multistage Sampling:** Combines multiple sampling methods (e.g., first cluster sampling, then random sampling within clusters).

Example:

Stage 1: Randomly select cities.

Stage 2: Randomly select neighborhoods within those cities.

Stage 3: Randomly select households within those neighborhoods.

B. Non-Probability Sampling Methods

In non-probability sampling, some elements of the population have no chance of being selected, or the probability of selection is unknown. These methods are often used in exploratory, qualitative, or resource-constrained studies.

1. **Convenience Sampling:**

Samples are selected based on ease of access. While quick and cost-effective, this method risks significant bias and limits generalizability.

2. **Judgmental or Purposive Sampling:**

Researchers use their expertise to choose participants who are deemed most useful or relevant. This method is common in case studies or expert panels.

3. **Quota Sampling:**

This involves selecting samples to fill predefined quotas for certain characteristics (e.g., gender or region), ensuring representation without random selection.

4. **Snowball Sampling:**

Existing participants refer new participants, creating a chain of referrals.

This approach is particularly helpful in studying hidden or hard-to-reach populations, such as specific online subcultures.

DESIGNING A SAMPLING STRATEGY IN DATA SCIENCE

A **sampling strategy** outlines the specific approach the researcher will use to select study participants or data entries. It must align with the research objectives, be practically feasible, and ethically sound.

Steps in Designing a Sampling Strategy:

1. **Define the Research Objectives:**

Clearly articulate what you aim to investigate. Are you exploring trends, comparing groups, or testing hypotheses?

2. **Identify the Target and Accessible Populations:**

Determine the ideal group to study and assess how much of it you can realistically reach.

3. **Choose an Appropriate Sampling Method:**

Decide whether a probability or non-probability method is most appropriate, considering the trade-offs between generalizability and feasibility.

4. **Determine Sample Size:**

Use statistical formulas or tools to calculate the necessary sample size. Consider desired confidence levels, margin of error, and population variability. Also account for expected non-responses.

5. **Develop a Recruitment and Data Access Plan:**

Describe how you will obtain permission, access the sample, and engage participants.

DATA COLLECTION METHODS IN DATA SCIENCE

Once a sample has been identified, the next crucial step is to collect data from that sample. The method chosen must be appropriate to the type of data needed, the research context, and the available resources.

Common Data Collection Methods:

1. Surveys and Questionnaires:

These tools are widely used to collect structured data on behaviors, opinions, or demographics. They can be administered online via platforms like Google Forms, SurveyMonkey, or Qualtrics, allowing for wide reach and efficient data handling.

2. Interviews (Structured and Semi-Structured):

Conducted in person or remotely, interviews allow for deeper insight into participant views. Structured interviews use a fixed set of questions, while semi-structured ones allow flexibility to explore themes as they arise.

3. Observations:

This method involves watching and recording behaviors or events as they occur. It can be passive (non-participatory) or active (the researcher interacts with the environment).

4. Web Scraping and APIs:

Data can be programmatically extracted from websites or accessed via APIs. These techniques are essential for real-time or large-scale data collection in data science.

5. Database Extraction:

Data is retrieved directly from existing databases using tools like SQL or MongoDB queries. This is common in studies that rely on enterprise or institutional datasets.

Choosing the Right Method:

Factors to consider include the type of data (qualitative vs. quantitative), the accessibility of participants or data points, and the need for confidentiality or informed consent.

ETHICAL AND INCLUSIVE SAMPLING PRACTICES

Ethical integrity is central to all phases of the research process, particularly when it involves human participants or sensitive data.

Key Ethical Principles:

- **Informed Consent:** Participants must be fully informed about the study's purpose, procedures, risks, and benefits, and voluntarily agree to participate.
- **Anonymity and Confidentiality:** Measures should be taken to protect the identities and data of participants.
- **Avoidance of Harm:** Researchers must ensure that no psychological, social, or physical harm befalls participants as a result of their involvement.
- **Transparency:** Researchers must be honest and open about how the data will be used and stored.

Inclusivity Considerations:

- Deliberately include underrepresented or marginalized groups.
- Avoid sampling strategies that exclude certain demographics unless scientifically justified.
- Use language and technology that accommodate diverse participants (e.g., multiple languages, accessible interfaces).
- Address algorithmic biases by ensuring diverse training data and testing for fairness.